

3.1.3 The development of species: evolution and classification

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the concepts of biological classification and species	To include the taxonomic ranks in the hierarchy of classification (domain, kingdom, phylum, class, order, family, genus, species) AND an outline of the biological and phylogenetic species concepts.
(b) the types of evidence used in biological classification and consideration of how theories change as new evidence is found	Evidence for hominid classification to include observable features (e.g. fossils) and molecular evidence (e.g. DNA). HSW7, HSW8
(c) the use of DNA barcoding in biological classification, examples of the genes used and consideration of the reasons for the choice of these genes	To include the use of mitochondrial genes (e.g. cytochrome c oxidase 1) in animals, and chloroplast genes in plants (no details of electrophoresis are required).
(d) the interpretation of phylogenetic trees and genetic data to show relatedness and classification	To include consideration of hominids, both extinct and extant, and hylobatids, including examples in which there is conflicting evidence. HSW5, HSW6, HSW8
(e) (i) behavioural, physiological and anatomical adaptations to the environment	To include the following adaptations in <i>Homo sapiens</i> : tool use and cultural adaptations for social bonding (behavioural), lactose tolerance and skin pigmentation (physiological), bipedalism and brain size (anatomical) AND adaptations of plants to their environment including adaptations to extremes of temperature, light and water.
(ii) practical investigation into adaptations of plants to environmental factors	

(f) the evolution of language as an example of a scientific question with many competing theories

To include discussion of why some scientific questions (e.g. “how did language evolve?”) are difficult to answer because of a lack of evidence, and consideration of competing theories (to include the “mother tongues” and “gossip” hypotheses).

HSW1, HSW2, HSW3, HSW7, HSW8

(g) adaptation and selection as components of evolution

To include the ideas of genetic variation, selection pressures and natural selection in relation to evolution.

(h) the definition and measurement of biodiversity

To include a consideration that biodiversity can exist at the genetic, species and ecosystem levels.

M0.2, M0.3, M0.4, M1.1

(i) the calculations of genetic diversity within populations.

To include the percentage of gene variants (alleles) in a genome.

proportion of polymorphic = $\frac{\text{number of polymorphic gene loci}}{\text{total number of loci}}$